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Short Communication

Operation of a bench-scale TiFe-based alloy tank under mild conditions for low-cost stationary hydrogen storage

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ABSTRACT

Activation processes of 55 kg TiFe-based alloy, filled in a conventional bench-scale tank (TiFe-based tank) under 1.0 MPa Gauge and at 80 °C, and its hydrogen absorption and desorption properties were studied. Following six ab/desorption cycles, the activation process was completed after the seventh absorption process. A long operating time (284 h) was necessary to complete the activation treatment. However, the operating time could be shortened by optimizing each activation process. At a constant hydrogen flow rate of 20 NL/min, the TiFe-based tank absorbed and desorbed >8 Nm^3 hydrogen at inlet brine temperatures of 60 °C and 70 °C, respectively. Heat exchange between the alloy bed and the brine circulation during desorption processes was about two times higher than during the absorption processes. Compared with 70 kg $LaNi_5$ filled in the same tank ($LaNi_5$ tank), the heat exchange was higher. This shows the TiFe-based tank would be more suitable for high hydrogen flow operations than the $LaNi_5$ tank.

Introduction

Hydrogen is a promising candidate for an energy carrier and a storage medium because of its portability and storability [1]. Compared with high-pressure hydrogen gas (>70 MPa) and liquid hydrogen, solid-state hydrogen storage in metal hydride (MH) is suitable for stationary on-site storage due to its high volumetric storage density and mild operation conditions. MH hydrogen storage has other advantages: safety, compactness, reliability, thermally driven storage and so on. Therefore, stationary hydrogen energy systems using MH have been developed [2].

In many cases, AB_5-type rare-earth based alloys, which are the most typical hydrogen storage alloys, e.g. $LaNi_5$ and $Mm(Ni, Al)_5$, have been used for bench-scale MH hydrogen storage for stationary systems (e.g. Refs. [3–6]), since these alloys have easy activation, high cycle stability, rapid kinetics, etc. However, the AB_5 alloys are expensive, and this prevents the wide practical use of MH hydrogen storage. Low-cost and non-rare-earth alloys are preferred.

The AB-type alloy TiFe [7] is a promising material for low-cost MH hydrogen storage because of the low cost of the starting materials, the abundance of resources and its high hydrogen capacity per unit volume. The main drawback of this alloy is that it is difficult to activate for hydrogen

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absorption. Until now, great efforts have been made to improve the activation of the binary alloy. For example, by substituting Fe for Mn (e.g. $TiFe_{0.8}Mn_{0.2}$) [8,9], or Ni ($TiFe_{0.9}Ni_{0.1}$) [10], setting compositions to be Ti-rich from the stoichiometric value [11] and so on [12,13]. These TiFe-based alloys can absorb hydrogen at near ambient conditions after mild activation treatments. Nevertheless, the activation conditions of the bench-scale experiment (e.g. >50 kg) are not the same at the lab-scale (e.g. <5 g). Further, high hydrogen pressure (e.g. > 3.0 MPa) and/or high temperature (e.g. > 200 °C) are needed [14,15]. Hence the structure and configuration of the MH tank is complicated and the manufacturing cost is expensive. For these reasons, the TiFe-based alloys are limited in their practical application.

If it were possible to perform activation at less than 1.0 MPa and 100 °C, the MH tank structure could be simple because hydrogen leaks could easily be prevented, and a conventional tank could therefore be used. Moreover, it could use water or a non-freezing solution as a heat exchange fluid. To cool and heat the alloy bed, heat exchangers are usually installed in the MH tank (see Fig. 1). If it were necessary to heat the alloy bed at temperatures above 100 °C, water (or solution) could not be used as the heat exchange fluid; oil or molten salt would have to be used, which would also lead to high costs [16].

Although operation under mild conditions, e.g. less than 1.0 MPa and 100 °C, is important for low-cost MH hydrogen storage, to the best of our knowledge, no previous study has been reported regarding the activation processes of the TiFe-based alloy (>50 kg); hence, there are still many uncertainties. In Japan, operation below 1.0 MPa Gauge (G) is absolutely necessary for non-high-pressure gas equipment because of the High Pressure Gas Safety Act [17].

Recently, 55 kg of TiFe-based alloy filled in the conventional tank (TiFe-based tank) has been successfully activated below 1.0 MPaG and 80 °C. In this short communication, we report the hydrogen ab/desorption properties as well as the activation processes in detail, and discuss the results in comparison with those of 70 kg $LaNi_5$ filled in the same MH tank ($LaNi_5$ tank). Few reports about the comparison between TiFe-based and $LaNi_5$ filled in the same bench-scale tank have been presented so far. We believe the obtained results in this paper are directly useful and important information for the wide and practical use of the TiFe-based alloy, which could lead to low-cost stationary MH hydrogen storage.

Experimental set-up

The TiFe-based alloy (Japan Metals & Chemicals Co., Ltd.) was $TiFe_{0.8}Mn_{0.2}$ with a few Mn substituting for V, and was prepared using a high-frequency melting furnace under an Ar atmosphere. The as-cast alloy was equilibrated by annealing for 24 h at 1100 °C under the Ar atmosphere. Its full hydrogen storage capacity, as measured by the hydrogen atom (H) to metal atom (M) ratio (H/M), was 0.9. The size of 50% of the grains was greater than 150 μm. The reaction heat for ab/desorption is 33.4 kJ/molH_2. The alloy (~5 g) was able to absorb hydrogen immediately after vacuuming for 1.5 h at 80 °C. Details of the alloy are described in Ref. [18].

Fig. 1(a) and (b) shows a photograph and schematic of the MH tank, respectively. This tank was of the conventional, vertical type. It was composed of a stainless steel cylindrical pipe and flanges [19,20]. The basic design requirements and its details are described elsewhere [21]. It had an internal diameter of 306 mm and a height of 700 mm. The tube heat exchanger to heat and cool the alloy bed, 50 U-shaped Cu circulation tubes, and seven hydrogen tubes with 0.5 micron-pore mesh vertically penetrated the alloy bed [see Fig. 1(a)]. Its weight and capacity were 70 kg and 20 L, respectively. 55 kg of the TiFe-base alloy and 70 kg of the $LaNi_5$ were placed in each MH tank. Both alloy beds were the same volume. Each MH tank was covered with a heat insulator and placed outside hydrogen storage units [19].

The heat exchange fluid was a 40% ethylene glycol (brine), which was used to prevent freezing, and its flow rate was 3.3 L/min. The brine temperature was controlled by a thermostatic

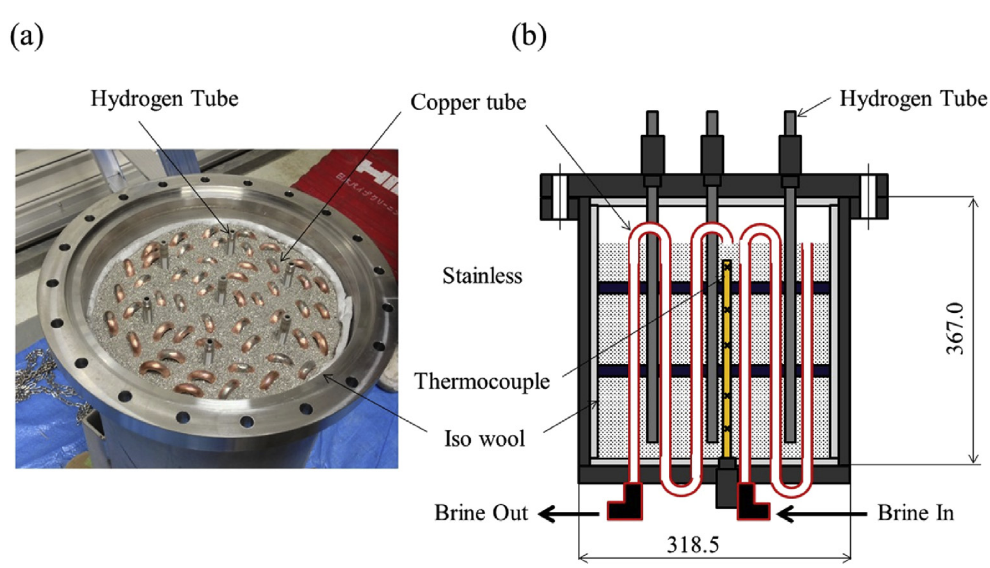

Fig. 1 – (a) Photograph and (b) schematic of the MH tank.

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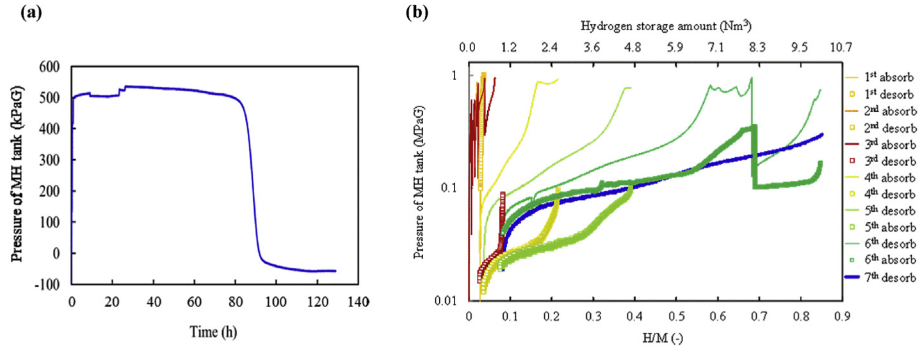

Fig. 2 – (a) Initial hydrogen absorption curve of the TiFe-based tank. (b) MH tank pressure profiles during the activation processes on a composition–pressure diagram.

controller (TOKYO RIKAKIKAI, NCC3000D, heat output; 1.6 kW). The heat exchange, q, between the brine circulation and the alloy bed was calculated using the following equation:

$$q = C_p \cdot \rho \cdot \Delta T \cdot v \tag{1}$$

where C_p is the specific heat of the circulation brine at constant pressure [3.85 kJ/(kg·K)], ρ is the density (1.02 kg/m^3), ΔT is the temperature difference between the inlet and outlet circulation brine [see Fig. 1(b)], and v is the flow rate of the circulation brine (3.3 L/min.).

Hydrogen was supplied from a water electrolyser under constant conditions at 900 kPaG. Mass flow controllers (HORIBA SEC-N100) measured and accurately controlled the hydrogen flow rate at 20 NL/min. The hydrogen flow rate under both absorption and desorption processes was the same. Desorbed hydrogen from the MH tank was emitted to the outside. The maximum hydrogen pressure was less than 1.0 MPaG to adhere to the High Pressure Gas Safety Act of Japan [17]. In the present study, we nominally chose gauge pressure to represent the pressure unit.

Results and discussion

Activation processes

Activation processes were carried out as follows: hydrogen was applied up to 800 kPaG after vacuuming for 10 min and this operation was cycled three times for impurity removal. Next, the alloy bed was heated up to 77 °C under a hydrogen pressure of 500 kPaG and kept at these conditions for 2 h, and then vacuumed [~50 Pa(abs.)] for 24 h at 77 °C. Hydrogen was applied up to 900 kPaG and kept for 24 h at the temperature to remove the alloy surface oxide (reduction treatment). Subsequently, three cycles of hydrogen application (~800 kPaG) and vacuuming (10 min) were carried out before the alloy bed was cooled down to 5 °C at 900 kPaG. We repeated these processes following the reduction treatment. The alloy bed was kept at a hydrogen pressure of 530 kPaG without any thermal controlling. The outside air temperature was 20–30 °C during this experiment.

Initial hydrogen absorption was observed after a long incubation time. Fig. 2(a) shows the initial hydrogen absorption curve of the TiFe-based tank. After being kept at 530 kPaG for 80 h, the pressure of the MH tank reduced. In the lab-scale experiment (<5 g), initial hydrogen absorption was observed immediately after 1 h vacuuming [~10^{-3} Pa(abs.)] at 80 °C; however, a long incubation time was needed in the bench-scale experiment. This is because the vacuum conditions ultimately created were quite different in the lab-scale and the bench-scale experiments.

After the initial hydrogen absorption, the absorption operation was carried out at 900 kPaG and 2 °C after vacuuming at 77 °C for 12 h. Low temperature absorption was chosen to make hydrogen absorption easier. Following this operation, each subsequent vacuuming and absorption time was 12 h. Fig. 2(b) shows the MH tank pressure profiles during the activation processes on a composition–pressure diagram. The second and third ab/desorption processes were carried out using the same methods and the third level of hydrogen storage was achieved by increasing step-by-step to H/M~0.1 so as not to exceed 1.0 MPaG [see Fig. 2(b)].

Both hydrogen ab/desorption temperatures were set at 30 °C after the third ab/desorption cycles to reduce the cooling and heating time. Hydrogen storage during the fourth stage reached H/M~0.2, i.e. 2.4 Nm^3 [the yellow line in Fig. 2(b)], which was due to the increase in the absorption rate achieved by increasing the temperature to 30 °C.

As shown by the yellow–green line in Fig. 2(b), by the fifth cycle, the hydrogen storage amount reached H/M~0.4 and had increased significantly compared to previous cycles. When repeating the ab/desorption cycles, cracking and pulverization occurred in the alloy particles, meaning the surface area was increased [1,18]. This is why this large increase in the hydrogen storage amount was observed.

By the sixth absorption process, indicated by the green line in Fig. 2(b), the hydrogen storage amount had continuously increased to H/M~0.58 at the hydrogen flow rate of 20 NL/min and at 30 °C. It was then increased step-by-step to H/M~0.69 so as not to exceed 1.0 MPaG. Hydrogen storage increased to H/M~0.86 at an alloy-bed temperature of 15 °C, which corresponds to a decrease in the absorption pressure at around H/M~0.69. This amount of hydrogen storage (H/M~0.86) almost represented full charge. The desorption pressure increased at around H/M~0.69 because the alloy-bed temperature was increased for a short time [represented by the green square in Fig. 2(b)].

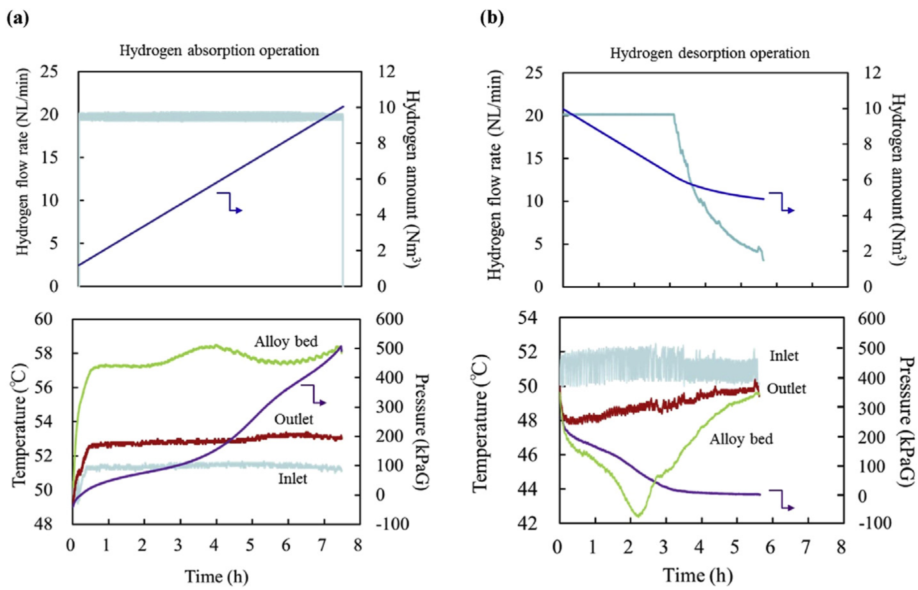

Fig. 3 – (a) Hydrogen flow rate and the hydrogen amount profiles (upper), and temperature and pressure profiles (lower) under hydrogen absorption operation condition and (b) under hydrogen desorption operation condition. Each inlet brine temperature was set at 50 °C.

As shown by the bold blue line in Fig. 2(b), the hydrogen storage amount continuously increased to H/M~0.86 at a constant hydrogen flow rate (20 NL/min), which shows that the activation processes were completed. The total operating time was 284 h. In the case of the $LaNi_5$ tank, initial hydrogen absorption was observed after 2 h vacuuming at 20 °C with no incubation time. The hydrogen storage amount continuously increased to H/M~0.85 by the fourth absorption process.

This long operating time under mild conditions was needed, however, it is possible to shorten this operating time by improving the vacuum level, and optimizing the vacuuming time and the reduction treatment. In particular, the vacuuming time was 12 h (one night), which was too long. Since the life cycle of hydrogen storage tanks using TiFe-based alloy has been reported to be over 7000 cycles, i.e. ~20 years, under practical operating conditions (ca. $0.15 < H/M < 0.85$; 1 cycle for 24 h), the activation operating time is negligible in comparison with the life cycle. Thus, we consider there to be merit to the initial cost reduction.

Hydrogen absorption and desorption properties

Hydrogen ab/desorption processes were operated using a hydrogen flow rate of 20 NL/min and the inlet brine temperature was set at 50, 60, and 70 °C. The upper panel of Fig. 3(a) shows the hydrogen flow rate and the hydrogen storage profiles, and the lower panel shows the temperature and pressure profiles during hydrogen absorption. In this part of the experiment, the inlet brine temperature was set at 50 °C.

Because of the constant hydrogen flow rate and the linear increase in the hydrogen storage amount, the TiFe-based tank constantly absorbed hydrogen. The absorption time was 7.5 h and the hydrogen storage amount was 8.5 Nm^3. Here, the hydrogen flow rate was 20 ± 0.4 NL/min due to the performance of the electrolyser. Given the reaction heat of hydrogen absorption, the outlet brine temperature increased by ~1 °C [see Fig. 3(a) lower]. The temperature of the alloy bed shows a broad peak toward high temperature at around 4 h operating time. This peak was a result of the increase in thermal conductivity due to the pressure increase caused by a phase transition from β hydride to γ hydride [22]. The outlet brine temperature showed no peak because the heat output of the thermostatic controller (1.6 kW) was much higher than the reaction heat of the alloy (483 W).

The hydrogen desorption process at an inlet brine temperature of 50 °C is shown in Fig. 3(b). As the upper panel of the figure shows, the hydrogen flow rate decreased immediately after 3.2 h of operation and the slope in the amount of hydrogen storage became gradual. Owing to the low equilibrium desorption pressure of the alloy (see Fig. 4), the amount of hydrogen desorbed under a constant hydrogen flow rate was about 4 Nm^3. Given the endothermic heat of hydrogen desorption, the outlet brine temperature decreased by ~2 °C for 3.2 h as shown in the lower panel of Fig. 3(b). This temperature difference, ΔT, was larger than that of the absorption process. Contrary to the absorption process, the temperature of the alloy bed shows a peak toward low temperature around at 2.2 h operating time. After that, both outlet brine and alloy bed temperatures increased, since the reaction heat reduced in accordance with the decrease in hydrogen flow rate. Fig. 4 shows the dynamic pressure-composition-temperature (PCT) isotherm diagram under the hydrogen flow rate of 20 NL/min, and Table 1 summarizes the hydrogen ab/desorption operations. This figure and table show that the inlet brine temperature needed to be set below 60 °C to absorb >8 Nm^3 hydrogen, and to be set above 70 °C to desorb >8 Nm^3 hydrogen. This performance mainly depended on the low and slope equilibrium pressure of the TiFe-based alloy.

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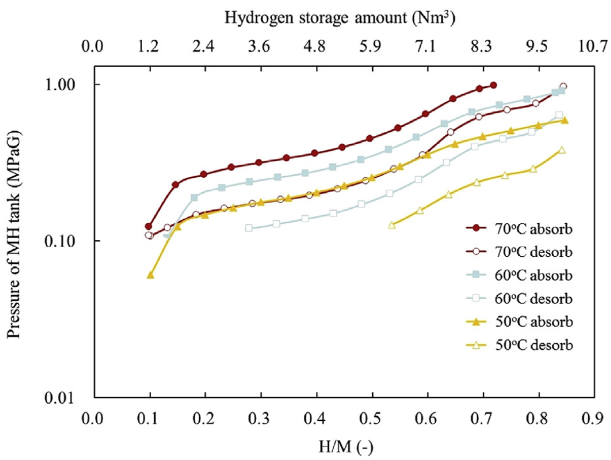

Fig. 4 – Dynamic pressure-composition-temperature (PCT) isotherm diagram of the TiFe-based tank under a hydrogen flow rate of 20 NL/min.

Here, we mention the heat exchange, q, as shown in Table 1. All q values under the desorption processes were about two times higher than for the absorption processes, which mainly depended on ΔT, as described in equation (1). The low value of ΔT under the absorption process was due to heat loss to the tank outside. Although each tank was covered with a heat insulator, we found out that the surface temperature of the heat insulator was obviously higher than the outside air temperature during the absorption process. Comparing the q values under the absorption processes, q decreased with increasing inlet brine temperature despite having the same hydrogen flow rate. This is because the heat loss at high temperature was larger than at low temperature.

Next we compare the results obtained from operating the TiFe-based tank with those of the $LaNi_5$ tank and discuss the operation of the former. The $LaNi_5$ tank was not able to absorb hydrogen at inlet brine temperatures >60 °C because of the high equilibrium absorption pressure, but was able to desorb 8 Nm^3 hydrogen at the inlet brine temperature of 50 °C; the latter tank is used to operate at the lower temperature range. Although it is preferable for the operation of the TiFe-based tank to prepare a heat source (>60 °C), we confirmed that a sufficient amount of heat was obtainable from solar thermal panels even in winter conditions [19] and from the waste heat of fuel cells [20]. This indicates that the preparation of the heat source is not so difficult and the TiFe-based tank is available for practical applications.

Table 2 shows the comparison of heat exchange, q, between the TiFe-based and $LaNi_5$ tank. The q of the former tank is higher than that of the latter. We consider this to be the effect of the heat conductivity rather than the reaction heat. Reaction heats and enthalpies for both tanks are similar, as presented in Table 2. In addition, the heat output of the thermostatic controller was much higher than their reaction heats. This means that the effect of the reaction heat would be small. On the other hand, $LaNi_5$ was considerably pulverized after the ad/desorption cycles and the void fraction of the alloy bed was increased [23]. In the TiFe-based alloy, such large pulverization did not occur [18]. The thermal conductivity of the alloy bed is greatly influenced by the void fraction and decreases with increases in the void fraction [23]; hence, a higher q was observed than in the $LaNi_5$ tank. This result indicates that the TiFe-based tank was more suitable for high hydrogen flow operations than the $LaNi_5$ tank. Details of the heat loss and its effects on the heat exchange and the hydrogen ab/desorption are currently being studied with the aid of a numerical simulation.

Table 2 – Comparison of heat exchange between TiFe-based and $LaNi_5$ tank under a hydrogen flow rate of 20 NL/min.

	TiFe-based tank	$LaNi_5$ tank
	Heat exchange, q (W)	
Absorption process	290	266
Desorption process	570	461
Reaction heat estimated from enthalpy	483	442
Enthalpy	31.8 (kJ/mol)[a]	30.1 (kJ/mol)[b]

[a] Estimated from von't Hoff plots.
[b] Literature value in Ref. [1].

Conclusions

We carried out activation treatments for 55 kg TiFe-based alloy filled in a conventional bench-scale tank (the TiFe-based tank) below 1.0 MPaG and 80 °C. The TiFe-based tank absorbed 8 Nm^3 under a constant hydrogen flow rate after the sixth absorption and desorption cycle. The long operating time for the activation processes could be shortened by improving the vacuum level and optimizing each process. We compared the hydrogen ab/desorption properties with those of 70 kg $LaNi_5$ filled in the same tank ($LaNi_5$ tank). The TiFe-based tank exhibited higher heat exchange values than the $LaNi_5$ tank. We therefore consider the former tank to be more suitable for operation under high hydrogen flow conditions

Table 1 – Summary of hydrogen storage amount and heat exchange under absorption and desorption processes.

Inlet brine temperature (°C)	Absorption process[a]		Desorption process[a]	
	H_2 amount (Nm^3)	Heat exchange, q (W)	H_2 amount (Nm^3)	Heat exchange, q (W)
70	7.3	216	8.3	550
60	8.5	261	6.5	585
50	8.8	290	4.0	570

[a] Hydrogen flow rate was 20 NL/min.

than the latter tank. The present study is relevant to the development of low-cost metal hydride stores using the TiFe-based alloy for stationary hydrogen storage.

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